

Topic 3: Asymptotics

02-680: Essentials of Mathematics and Statistics

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1 Big O

To understand function behavior of say function $f(x)$ we often want to look at its growth compared to another function $g(x)$, for instance we might ask:

Does $f(x)$ grow no faster than $g(x)$?

Define the **set** of functions $f(x)$ for which this is true:

For functions $h : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ and $g : \mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$

$$O(g(x)) := \left\{ h(x) \mid \exists x_0 \geq 0, c > 0 : \forall x \geq x_0 : h(x) \leq c \cdot g(x) \right\}$$

(notice the explicit use of $\geq$ vs $>$)

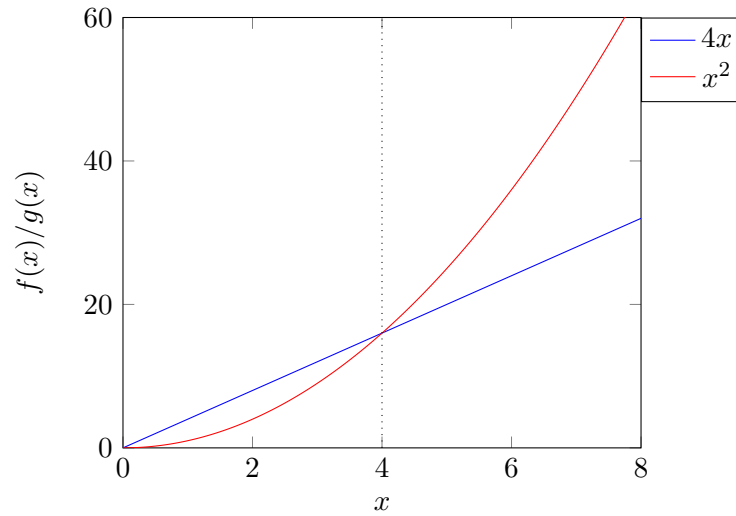
The question above then becomes

Is $f(x) \in O(g(x))$?

pedantry detail

You might see someone use $f(x) = O(g(x))$, this is not technically correct, as its asking if a function is equivalent to a set of functions.

So as an example: $f(x) := 4x$ and $g(x) = x^2$.



- $f(x) \in O(g(x) + h(x)) \iff f(x) \in O(\max(g(x), h(x)))$
- $f(x) \in O(g(x))$ and $g(x) \in O(h(x)) \implies f(x) \in O(h(x))$
- $f(x) \in O(h_1(x))$ and $g(x) \in O(h_2(x)) \implies f(x) + g(x) \in O(h_1(x) + h_2(x))$
- $f(x) \in O(h_1(x))$ and $g(x) \in O(h_2(x)) \implies f(x) \cdot g(x) \in O(h_1(x) \cdot h_2(x))$

1.1 Polynomials

Let $p(x) = \sum_{i=0}^k a_i x^i$ be a polynomial of order k , $p(x) \in O(x^k)$.

Basic idea of proof: for $i < k$, $a_i x^i < |a_i| x^k$ for any $x > 1$.

1.2 Logarithms & Exponentials

Log « Polynomial « Exponential

$\log x \in O(n^\epsilon)$ for $\epsilon > 0$. (base of the log does not matter)

$p(x) \in O(b^x)$ for any $b > 1$ and $k \geq 0$ for the polynomial above.

$b^x \in O(c^x) \iff b \leq c$.

1.3 Useful Examples

Is $3x^2 + 2 \in O(x^2)$? Is $x^3 \in O(x^2)$?

Is $3x + 4y + 5xy \in O(x)$? Is it $\in O(y)$? Is it $\in O(xy)$?

2 Others

$$\Omega(g(x)) := \left\{ h(x) \mid \exists x_0 \geq 0, d > 0 : \forall x \geq x_0 : h(x) \geq d \cdot g(x) \right\}$$

$$f(x) \in O(g(x)) \iff g(x) \in \Omega(f(x))$$

$$\Theta(g(x)) := \left\{ h(x) \mid h(x) \in O(g(x)) \wedge h(x) \in \Omega(g(x)) \right\}$$

$$o(g(x)) := \left\{ h(x) \mid \forall c > 0 : \exists x_0 \geq 0 : \forall x \geq x_0 : h(x) < c \cdot g(x) \right\} := \left\{ h(x) \mid h(x) \in O(g(x)) \wedge h(x) \notin \Omega(g(x)) \right\}$$

$$\omega(g(x)) := \left\{ h(x) \mid \forall c > 0 : \exists x_0 \geq 0 : \forall x \geq x_0 : h(x) > c \cdot g(x) \right\} := \left\{ h(x) \mid h(x) \notin O(g(x)) \wedge h(x) \in \Omega(g(x)) \right\}$$

2.1 Useful Example

What is the relationship between these two functions:

$$f(x) = x^2 \quad g(x) = \begin{cases} x^3 & x \% 2 = 0 \\ x & x \% 2 = 1 \end{cases}$$

Is $f(x) \in O(g(x))$? Is $f(x) \in \Omega(g(x))$? Is $f(x) \in o(g(x))$? Is $f(x) \in \omega(g(x))$?

Useful References

Liben-Nowell, “Connecting Discrete Mathematics and Computer Science, 2e”. §6.2